

CS 311: Parallel Computing Lab

Lab Assignment #1

Due Date: Sep 1, 2026

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Write parallel and sequential programs for following tasks. Execute the code multiple times and compare the execution times of sequential and parallel programs. Use an array of large size (say $n=2^{20}$) of randomly generated numbers, wherever required.

1. Multiplication of two one dimensional arrays of size n .
2. Average of n numbers.
3. Minimum of n numbers.
4. Factorial value of an integer.

Submission: The programs written for these problems must be submitted at the end of the lab. Submit your answers through **LMS (Moodle)** by uploading the required file(s).

Please name your files according to the question number, as follows:

Question 1: Question_1_Ans

Question 2: Question_2_Ans

Question 3: Question_3_Ans

Question 4: Question_4_Ans